Causal Research Design

Block 6

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Meaning of Causality

Ordinary Meaning

- X is the only cause of Y
- X must always lead to Y
- It is possible to prove that X is a cause of Y

Scientific Meaning

- X is only one of a number of possible causes of Y
- The occurrence of X makes the occurrence probable (X is a probabilistic cause of Y)
- We can never prove that X is a cause of Y At best, we can infer that X is a cause of Y

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- Concomitant variation is the extent to which a cause, X and an effect, Y occur together or vary together in the way predicted by the hypothesis under consideration.
- The time order of occurrence condition states that the causing event must occur either before or simultaneously with the effect; it cannot occur afterwards.
- The absence of other possible causal factors means that the factor or variable being investigated should be the only possible causal explanation.

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Types of Variables

- **Independent variables** are variables or alternatives which are manipulated and whose effects are measured and compared – for example, price levels.

- **Dependent variables** are the variables which measure the effect of the independent variables on the test units – for example, sales, profits and market shares.

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Test Units

- Test units are individuals, organizations or other entities whose response to the independent variables or treatments is being examined

- For example, consumers or stores.

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Experimental Design

An experimental design is a set of procedures specifying:

- the test units and how these units are to be divided into homogeneous subsamples
- which independent variables or treatments are to be manipulated
- which dependent variables are to be measured
- how the extraneous variables are to be controlled.

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Symbols Used

We define the following set of symbols commonly used in market research:

- X=The exposure of a group to an independent variable, treatment or event, the effects of which are to be determined
- O= The process of observation or measurement of the dependent variable on the test units or group of units
- R= The random assignment of participants or groups to separate treatments.

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Validity

- **Internal validity** refers to whether the manipulation of the independent variables or treatments actually caused the observed effects on the dependent variables. Control of extraneous variables is a necessary condition for establishing internal validity.

- **External validity** refers to whether the cause-and-effect relationships found in the experiment can be generalised. To what populations, settings, times, independent variables and dependent variables can the results be projected?

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Extraneous Variables

- Extraneous variables are all variables other than the independent variables which affect the response of the test units

- For example, store size, store location and competitive effort.

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Extraneous Variables Types

- **History** (H) refers to specific events which are external to the experiment but occur at the same time as the experiment.
- **Maturation** (MA) refers to changes in the test units themselves which occur with the passage of time.
- **Instrumentation** (I) refers to changes in the measuring instrument, in the observers or in the scores themselves.
- **Statistical Regression** (SR) effects occur when test units with extreme scores move closer to the average score during the course of the experiment.
- **Selection Bias** (SB) refers to the improper assignment of test units to treatment conditions.
- **Mortality** (MO) refers to the loss of test units while the experiment is in progress.

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Extraneous Variables: Testing Effects

- Testing effects are caused by the process of experimentation.
- Typically, these are the effects on the experiment of taking a measure on the dependent variable before and after the presentation of the treatment.
- The main testing effect (MT) occurs when a prior observation affects a later observation. In the interactive testing effect (IT), a prior measurement affects the test unit's response to the independent variable.

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Controlling Extraneous Variables

- **Randomization** refers to the random assignment of test units to experimental groups by using random numbers. Treatment conditions are also randomly assigned to experimental groups.

- **Matching** involves comparing test units on a set of key background variables before assigning them to the treatment conditions.

- **Statistical Control** involves measuring the extraneous variables and adjusting for their effects through statistical analysis.

- **Design Control** involves the use of experiments designed to control specific extraneous variables.

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What We Want to Isolate

- In any research we want to isolate the treatment effect (TE)
- This is considered the causal effect of the treatment

Pre Expiremental Designs

Pre-experimental designs do not employ randomization procedures to control for extraneous factors:

- the one-shot case study
- the one-group pre-test-post-test design
- the static group.

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True Experimental Design

In true experimental designs, the researcher can randomly assign test units to experimental groups and treatments to experimental groups:

- The pre-test-post-test control group design
- The post-test-only control group design
- The Solomon four-group design.

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Quasi-Experimental Design

Quasi-experimental designs result when the researcher is unable to achieve full manipulation of scheduling or allocation of treatments to test units but can still apply part of the apparatus of true experimentation:

- Time series
- Multiple Time Series Designs.

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Statistical Designs

Statistical designs are a series of basic experiments which allows for statistical control and analysis of external variables:

- Randomized Block Designs
- Latin Square Designs
- Factorial Designs.

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One-Shot Case Study

X_1 O_1

A single group of test units is exposed to a treatment X

A single measurement on the dependent variable is taken (O_1).

There is no random assignment of test units.

The one-shot case study is more appropriate for exploratory research than for conclusive research.

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One-Group Pre-Test-Post-test design

O_1 X O_2

A group of test units is measured twice. There is no control group.

The treatment effect is computed as $O_2 - O_1$

. The validity of this conclusion is questionable since extraneous variables are largely uncontrolled.

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Static Group

EG: X $O_1$and. CG: O_2

A two-group experimental design.

The experimental group (EG) is exposed to the treatment, and the control group (CG) is not.

Measurements on both groups are made only after the treatment. Test units are not assigned at random.

The treatment effect would be measured as $O_1 - O_2$

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Pre-Test-Post-Test Control Group Design

EG: $R\ O_1\ X\ O_2$ and CG: $R\ O_3\ O_4$.

Test units are randomly assigned to either the experimental or the control group.

A pre-treatment measure is taken on each group.

The treatment effect (TE) is measured as:

$$(O_2 - O_1) - (O_4 - O_3).$$

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Pre-Test-Post-Test
Controlling Extraneous Factors

Selection bias is eliminated by randomization. The other extraneous effects are controlled as follows:

$$O_2 - O_1 = \text{TE+H+MA+MT+IT+I+SR+MO}$$

And

$$O_4 - O_3 = \text{H+MA+MT+I+SR+MO.}$$

Therefore

$$(O_2 - O_1) - (O_4 - O_3) = \text{TE+IT.}$$

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Post-Test-Only Control Group Design

EG: $R\ X\ O_1$ and CG: RO_2

The treatment effect is $O2-O1$=TE

Except for pre-measurement, the implementation of this design is very similar to that of the pre-test-post-test control group design.

It is assumed that the two groups are similar in terms of pre-treatment measures on the dependent variable because of the random assignment of participants to groups

sensitive to selection bias and mortality

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Quasi-Expiremental Design

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Time Series Design

A time series design involves a series of periodic measurements:

$$O_1\ O_2\ O_3\ O_4\ O_5\ O_6\ O_7\ O_8\ O_9\ O_{10}$$

There is no randomisation of test units to treatments. The timing of treatment presentation, as well as which test units are exposed to the treatment, may not be within the researcher's control.

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Multiple Time Series Design

In a multiple time series design, we have:

EG: O_1 O_2 O_3 O_4 O_5 X O_6 O_7 O_8 O_9 O_{10}

CG: O_{11} O_{12} O_{13} O_{14} O_{15} O_{16} O_{17} O_{18} O_{19} O_{20}

The improvement lies in the ability to test the treatment effect twice:

- against the pre-treatment measurements in the experimental group
- against the control group

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Statistical Designs

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Advantages of Statistical Designs

Statistical designs consist of a series of basic experiments which allow for statistical control and analysis of external variables and offer the following advantages.

- The effects of more than one independent variable can be measured.
- Specific extraneous variables can be statistically controlled.
- Economical designs can be formulated when each test unit is measured more than once.

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Randomized Block Design

- is useful when there is only one major external variable, such as sales or store size, which might influence the dependent variable
- The test units are blocked, or grouped, on the basis of the external variable.
- By blocking, the researcher ensures that the various experimental and control groups are matched closely on the external variable

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Randomized Block Design Example

		Treatment groups		
Block number	**Mercedes usage**	*Advertisement A*	*Advertisement B*	*Advertisement C*
1	*High*	A	B	C
2	*Medium*	A	B	C
3	*Light*	A	B	C
4	*None*	A	B	C

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Latin Square Design

- A Latin square design allows the researcher to control statistically two non-interacting external variables as well as to manipulate the independent variable
- Each external or blocking variable is divided into an equal number of blocks, or levels. The independent variable is also divided into the same number of levels
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- The independent variable is also divided into the same number of levels

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Latin Squares Examples

	Interest in watching Formula One races		
Mercedes usage	*High*	*Medium*	*Low*
High	B	A	C
Medium	C	B	A
Light and none	A	C	B

Note: A, B and C denote three test advertisements, which have, respectively, no humour, some humour and high humour.

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Factorial Design

- is used to measure the effects of two or more independent variables at various levels.
- factorial designs allow for interactions between variables.

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Factorial Design Example

Amount of Mercedes E-Class information	**Amount of humour**		
	No humour	*Some humour*	*High humour*
Low	A	B	C
Medium	D	E	F
High	G	H	I

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Limitations of Experiences

- Experiments can be time consuming, particularly if the researcher is interested in measuring the long-term effects.
- Experiments are often expensive. The requirements of experimental group, control group and multiple measurements significantly add to the cost of research.
- Experiments can be difficult to administer. It may be impossible to control for the effects of the extraneous variables, particularly in a field environment.
- Competitors may deliberately contaminate the results of a field experiment.

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